

DS1112SG60 diode stack at 13.8 kV: how the PLECS and Simscape diode parameters are obtained from the datasheet

Plain-English version of the calculation note “Parametri PLECS e Simscape del DS1112SG60”, Rev. 01, 2026-10-01 (Secom Proposal Engineering, D. Bagnara)

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The original Italian note is kept next to this file: [original/plecs_diode_parameters_ds1112sg.pdf](#). This version keeps every number, formula, table and conclusion of the original. Only the words are simpler. Where a sentence of the original packs several ideas together, it is split into short sentences here. Nothing new is added; a few short “in other words” remarks are marked as such.

If the physics behind chapter 5 (stored charge, lifetime, why the recovery looks the way it does) is the hard part, read the companion document [physics_of_reverse_recovery.md](#) first or alongside.

Revision history of the original note

Rev.	Date	Author	Description
00	2026-10-01	D. Bagnara	First issue. How the ten parameters of the PLECS initialisation block are derived from the datasheet DS4181-5.0 and from the Dynex application note AN6531. Operating point of the B6 bridge at 13.8 kV, 60 A.
01	2026-10-01	D. Bagnara	Added section 7: the same data converted into the parameters of the charge-controlled diode (Lauritzen–Ma) of the MATLAB/Simscape model (<code>diode_lm_params.m</code>), a table of the twelve Dynex/Infineon sets, and a check on the test circuit.

Symbols used in this note

Symbol	Meaning
I_{F0} , <code>If0_ref</code>	forward current of the diode just before it turns off (the reference current of the model)
$a = dI_r/dt$, <code>di_dt_ref</code>	slope of the diode current while it is falling to zero, forced by the circuit (A/s)
Q_{rr}	reverse recovery charge: the area under the negative part of the current (C)
I_{rrm}	peak of the reverse current (A)

Symbol	Meaning
t_{rr}	reverse recovery time (s)
t_a, t_b	time from the zero crossing to the peak, and from the peak back to zero; $t_{rr} = t_a + t_b$
$s = t_b/t_a$	softness factor
τ	carrier lifetime (s)
T_M	transit time of the charge-controlled model (s)
V_{RRM}	repetitive peak reverse voltage rating
V_{TO}, r_T	threshold voltage and slope resistance of the forward characteristic (datasheet)
V_{f0}, r_t	the same two numbers as the PLECS parameters <code>Vf0</code> , <code>rt</code>
R_{off}	resistance of the model when the diode is blocking
C_j	junction capacitance
T_{vj}, T_j	junction temperature
$n = 6$	number of diodes in series in one stack
C_{sn}, R_{sn}	snubber capacitor and resistor across each diode
R_b	static balancing resistor across each diode
L_c	commutation inductance on each AC phase
$u, \text{Max_Qrr_unbalance}$	relative spread of Q_{rr} between the diodes of a stack

1 Goal and result

The PLECS model of the rectifier (`mv_dstack_six_pulse_rectifier.plecs`) uses the component *Diode with Reverse Recovery* for every diode of the stack. Its parameters are set in the initialisation script.

This note shows, one parameter at a time, how the values of the block below come from three sources:

- the Dynex DS1112SG datasheet (DS4181-5.0) [1];
- the Dynex application note AN6531 [2];
- the operating point of the circuit.

It also says, for each value, whether it is read directly from the datasheet, extrapolated from it, or only estimated.

Section 7 then shows how the same data become the physical parameters (τ , T_M , I_s , R_s) of the charge-controlled diode used in the MATLAB/Simscape twin model. It also checks that the two models recover the same charge with the same peak current.

```
Vrrm=6000;          %V   - DS1112SG60 (VRSM 6100)
roff=1.2e6;          %Ohm - 3253 V / 2.7 mA at Tj=100 C, law 0.96
                    %     (2.1e6 with the Dynex curves; 8e4 at VRRM and 150 C)
Cj=5e-10;            %F   - estimate, not in the datasheet
Vf0=0.9;             %V   - VT0 max at 150 C
rt=0.93e-3;          %Ohm - rT max at 150 C
If0_ref=60;          %A
di_dt_ref=1.07e5;    %A/s - V0/(2Lc)
Qrr_ref=0.45e-3;     %As  - middle value at 100 C (floor 0.32e-3 / 7.0 A / 92 us;
                    %     worst 1.2e-3 / 13.4 A / 176 us)
Irrm_ref=8.3;        %A
Trr_ref=2*Qrr_ref/Irrm_ref; %s - 109 us
```

Table 1 says where each value comes from. Three parameters are read directly from the datasheet (V_{RRM} , V_{TO} , r_T). Two come from the circuit (I_{F0} , dI_r/dt). Three are extrapolations of the datasheet to the operating point (R_{off} , Q_{rr} , I_{rrm} ; t_{rr} follows from them). One is an estimate (C_j).

Table 1: where the parameters of the PLECS block come from.

Parameter	Value	Origin	Type
Vrrm	6000 V	<i>Voltage ratings</i> table, grade SG60	datasheet value
Vf0	0.9 V	V_{TO} max at $T_{vj} = 150$ °C	datasheet value
rt	0.93 mΩ	r_T max at $T_{vj} = 150$ °C	datasheet value
roff	1.2 MΩ	$I_{RM} = 75$ mA at V_{RRM} , 150 °C, moved to 3253 V, 100 °C	extrapolation (§4.3)
Cj	0.5 nF	not in the datasheet; estimated depletion capacitance	estimate (§4.4)
If0_ref	60 A	DC current of the bridge	circuit (§3)
di_dt_ref	1.07×10^5 A/s	$V_0/(2L_c)$ at the end of the commutation	circuit (§3)
Qrr_ref	0.45 mC	Q_S of datasheet Fig. 4 extrapolated to 0.107 A/μs and 100 °C	extrapolation (§5.2)
Irrm_ref	8.3 A	from the recovery triangle with Q_{rr} , dI_r/dt and softness 0.4	derived (§5.3)
Trr_ref	109 μs	$2Q_{rr}/I_{rrm}$	derived (§5.3)

2 The PLECS model and what it asks for

The PLECS component *Diode with Reverse Recovery* [3] is an ideal diode made of straight pieces (V_f , R_{on} , R_{off}). On top of it, a small behavioural network adds the reverse recovery:

- a small measuring inductance L_{rr} ;
- a resistor;
- a current source, driven linearly by the voltage across L_{rr} . That voltage is proportional to the slope di/dt of the diode current.

The recovery parameters are:

- the reference operating point, I_{F0} and dI_r/dt ;
- three sizes of the recovery triangle: t_{rr} , I_{rrm} and Q_{rr} .

Only two of the three triangle sizes are independent. If you give all three, PLECS ignores Q_{rr} and uses t_{rr} and I_{rrm} , with $Q_{rr} = \frac{1}{2} I_{rrm} t_{rr}$.

Two practical consequences decide which values to write.

1. **The model scales with the real di/dt .** The current source is proportional to $L_{rr} di/dt$. So the recovery current produced in the circuit is roughly

$$I_{rrm} \cdot \frac{(di/dt)_{\text{circuit}}}{(dI_r/dt)_{\text{parameter}}}.$$

Take the numbers of a datasheet measured at 10 A/μs and put them in a circuit that runs at 0.1 A/μs. The model then extrapolates over two decades. It produces a recovery one hundred

times smaller than the nominal one, and nothing warns you. This is why the parameters must be given *at the operating point of the circuit*: dI_r/dt equal to the real slope, and I_{F0} equal to the real current. Then the scaling ratio is one, and the simulated recovery is exactly what is written in the parameters.

2. **Geometric constraint** $I_{rrm} < t_{rr} dI_r/dt$. The current reaches its peak along the ramp, after $t_a = I_{rrm}/(dI_r/dt)$. The peak must fall inside t_{rr} (Fig. 1). If the condition is not met, the script stops with the message *Parameters for reverse recovery must satisfy: $I_{rrm} < t_{rr} * dI_r/dt$* .

Figure 1 (described). The recovery triangle used by PLECS. The current i_D starts at I_F and falls with the slope $-dI_r/dt$ forced by the circuit. It crosses zero at t_0 . It keeps falling with the same slope and reaches $-I_{rrm}$ after $t_a = I_{rrm}/(dI_r/dt)$. Then it goes back to zero in a straight line, in a time t_b . The whole recovery lasts $t_{rr} = t_a + t_b$. The grey triangle under the zero line has area $Q_{rr} = \frac{1}{2}I_{rrm}t_{rr}$. The ratio $s = t_b/t_a$ is the softness factor. From $Q_{rr} = \frac{1}{2}I_{rrm}t_{rr}$ and $t_{rr} = (1 + s)t_a$ it follows that

$$I_{rrm} = \sqrt{\frac{2Q_{rr}(dI_r/dt)}{1+s}}.$$

The same triangle, drawn next to the charge-controlled waveform, is [figure 2 of the physics document](#).

3 Operating point of the circuit: I_{F0} and dI_r/dt

In the B6 bridge with an inductive load, each diode conducts for 120°. It turns off at the end of the commutation with the diode of the next phase. During the commutation, the current of the outgoing diode falls with a slope set by the instantaneous line-to-line voltage applied to the two commutation inductances in series, $2L_c$.

Let $\hat{V} = \sqrt{2}V_{LL}$ be the peak line-to-line voltage, $\omega = 2\pi f$ and I_d the DC current. The overlap angle μ , the commutation voltage V_0 at the end of the overlap, and the slope at the zero crossing of the current are:

$$\cos \mu = 1 - \frac{2\omega L_c I_d}{\hat{V}}, \quad V_0 = \hat{V} \sin \mu, \quad \left. \frac{di}{dt} \right|_{i=0} = \frac{V_0}{2L_c}. \quad (3.1)$$

With the data of the script ($V_{LL} = 13.8$ kV, $f = 50$ Hz, $I_d = 60$ A, $L_c = L_{line} = 31.1$ mH, which is a 1.1 MVA transformer with $u_k = 6$ %):

$$\hat{V} = 19.52 \text{ kV}, \quad \mu = 20.0^\circ, \quad V_0 = 6.66 \text{ kV}, \quad \frac{di}{dt} = \frac{6660}{2 \times 0.0311} = 1.07 \times 10^5 \text{ A/s} = 0.107 \text{ A/}\mu\text{s}.$$

This is `di_dt_ref`.

The ramp lasts $I_d/(di/dt) = 560$ μs . That is much longer than the carrier lifetime (30–100 μs). So this is a *long ramp*. This fact decides how Q_{rr} is extrapolated (§5.2).

`If0_ref` is the real forward current, 60 A, not the 1000 A of the datasheet. In the long-ramp regime the charge left at the zero crossing does not depend on I_F . So the model must not rescale anything with respect to the reference current.

With the 1.25 MVA transformer ($L_c = 29$ mH) the slope is 0.114 A/ μs ; with $u_k = 4$ % it is 0.14 A/ μs . The slope must be recalculated if the transformer changes.

4 Static parameters

4.1 V_{RRM}

The datasheet lists voltage grades from SG55 (5500 V) to SG60 (6000 V), with $V_{RSM} = V_{RRM} + 100$ V. The design assumes grade SG60, which must be written in full in the order: **Vrrm = 6000**.

In PLECS this value is only a reference. The ideal diode does not break. So the check of the peak voltage on the first diode against $0.8 V_{RRM} = 4800$ V must be done afterwards on the waveforms.

4.2 V_{f0} and r_t

The forward characteristic is given as a threshold plus a slope: $v_F = V_{TO} + r_T i_F$, with $V_{TO} \leq 0.9$ V and $r_T \leq 0.93$ m Ω at $T_{vj} = 150$ °C. So **Vf0 = 0.9**, **rt = 0.93e-3**.

This threshold model is fitted for currents of hundreds of amperes. At 60 A it overestimates the voltage drop (the equation of datasheet Fig. 2, valid from 500 A, would give about 0.9 V in total at 60 A and 125 °C). So the forward loss

$$P_F = V_{TO} I_{F(AV)} + r_T I_{F(RMS)}^2 = 0.9 \times 21 + 0.93 \times 10^{-3} \times 33^2 = 19.9 \text{ W}$$

is conservative. For the sizing of the stack, V_{f0} and r_t matter little. They set the losses in the diodes and the voltage at which the level is clamped during conduction.

4.3 R_{off} : from the rated reverse current to the operating point

The datasheet gives one number only: $I_{RM} \leq 75$ mA at V_{RRM} and $T_{case} = 150$ °C. That is $R_{off} = 6000/0.075 = 80$ k Ω at the test point. It is a production test limit, not the value at the operating point, where both voltage and temperature are lower.

There are no curves for the DS1112SG. So the general curves of application note AN6531 [2] are used (its Fig. 4, taken from AN6161). The manufacturer itself calls them approximate.

- **Voltage:** $I_R(V)/I_R(V_{RRM}) \approx 0.30$ at 54 % of the rated voltage. The peak voltage per level is $\hat{V}/6 = 3253$ V, which is 54 % of 6000 V.
- **Temperature:** the curve, normalised at 125 °C, is about 0.30 at 100 °C, 1.0 at 125 °C, 2.7 at 140 °C. Extending it with the same exponential trend (doubling every ≈ 13 K) gives ≈ 4.2 at 150 °C, which is the temperature of the datasheet value. So the factor between 150 and 100 °C is $0.30/4.2 = 0.07$.

As an alternative, one can use the law that Infineon gives for the D711N65T, $0.96^{(T_{vj,max}-T_j)}$, which doubles every 17 K. Between 150 and 100 °C it gives 0.13. It is less steep, so more conservative. Table 2 gives the reverse current of a device at the limit, and the corresponding R_{off} , with the two laws.

The value written in the script, **roff = 1.2e6**, is the one of the $0.96^{\Delta T}$ law at $T_j = 100$ °C. It is the more conservative of the two at that temperature.

In the model, R_{off} acts only on the static voltage sharing (in parallel with R_b) and on the reverse losses. To simulate the leakage unbalance, the “best” diode gets a very high R_{off} (20–30 M Ω) and the other five get the value of the table. A typical device leaks three to ten times less than the limit.

Table 2: reverse current of a device at the limit ($I_{RM} = 75$ mA at 6000 V, 150 °C), moved to 3253 V, and $R_{off} = 3253/I_R$.

T_j	I_R , Dynex curves	R_{off}	I_R , law $0.96^{\Delta T}$	R_{off}
70 °C	0.53 mA	6.1 MΩ	0.80 mA	4.1 MΩ
85 °C	0.80 mA	4.1 MΩ	1.48 mA	2.2 MΩ
100 °C	1.53 mA	2.1 MΩ	2.73 mA	1.2 MΩ
110 °C	2.62 mA	1.2 MΩ	4.10 mA	0.8 MΩ
125 °C	5.87 mA	0.55 MΩ	7.57 mA	0.43 MΩ
150 °C	22.5 mA	0.14 MΩ	21 mA	0.15 MΩ

4.4 C_j

The datasheet does not give the junction capacitance. It is estimated as the depletion capacitance of a wafer of 40 mm diameter ($A \approx 1.3 \times 10^{-3} \text{ m}^2$) with a depleted zone of 0.3–0.5 mm at 3 kV:

$$C_j = \frac{\varepsilon_{Si} A}{w} \approx 0.3\text{-}0.5 \text{ nF}, \quad \text{so } C_j = 5\text{e-}10.$$

This is 0.5 % of $C_{sn} = 100 \text{ nF}$. It does not affect the voltage sharing or the losses. Its only job is to give the ideal diode a finite capacitance, for the numerical stability of the fast edges (corner frequency $1/(2\pi R_{sn} C_j) \approx 1.4 \text{ MHz}$).

5 Recovery parameters

5.1 The rated point and Fig. 4 of the datasheet

The datasheet characterises the recovery at one point only:

- $Q_S = 3000 \text{ }\mu\text{C}$ and $I_{rr} = 90 \text{ A}$,
- at $I_F = 1000 \text{ A}$, $dI_{RR}/dt = 3 \text{ A}/\mu\text{s}$, $T_{case} = 150 \text{ }^\circ\text{C}$, $V_R = 100 \text{ V}$.

Fig. 4 of the datasheet extends Q_S as a function of the slope, from about $0.7 \text{ A}/\mu\text{s}$ ($\approx 2100 \text{ }\mu\text{C}$) to $60 \text{ A}/\mu\text{s}$ ($\approx 6500 \text{ }\mu\text{C}$). On a log-log plot it is a straight line with slope

$$Q_S \propto \left(\frac{di}{dt}\right)^{0.28}, \quad \frac{\log(6500/3000)}{\log(50/3)} \approx 0.28. \quad (5.1)$$

In other words: the charge grows with the slope, but much more slowly than in proportion. Ten times the slope gives only about twice the charge.

There is no curve of I_{rr} . It is rebuilt from the triangle (§5.3). At the rated point: $t_a = 90/3 = 30 \text{ }\mu\text{s}$, $t_{rr} = 2 \times 3000/90 = 67 \text{ }\mu\text{s}$, so $t_b = 37 \text{ }\mu\text{s}$ and softness $s = t_b/t_a = 1.2$.

5.2 Why Q_{rr} at the operating point cannot be read, and how it is extrapolated

Our circuit works at $0.107 \text{ A}/\mu\text{s}$, seven times below the last published point. The physics of the long ramp says how to extrapolate.

The stored charge q obeys

$$\frac{dq}{dt} = i - \frac{q}{\tau},$$

where τ is the carrier lifetime. For a ramp $i = I_F - a t$ that lasts much longer than τ , the charge follows the current with a constant delay: $q \approx \tau i(t) + a\tau^2$. At the zero crossing of the current there is still

$$q(t_0) \approx a\tau^2, \quad (5.2)$$

which does not depend on I_F and is linear in the slope a .

In other words: the charge inside the diode cannot follow the current instantly. It lags behind by one lifetime. When the current reaches zero, the charge is still what the current had one lifetime earlier, $a\tau$, multiplied by τ . The physics document explains this step in detail.

From the rated point one gets an *apparent* lifetime $\tau = \sqrt{Q_S/a} = \sqrt{3 \times 10^{-3}/3 \times 10^6} = 32 \mu\text{s}$ at 3 A/ μs , and 55 μs at 0.7 A/ μs . The apparent lifetime grows as the slope decreases. The reason: at high di/dt a part of the charge recombines during the recovery and is never measured at the terminals. When the ramp becomes long, the apparent lifetime saturates towards the true lifetime of the wafer (50–100 μs for a 6 kV diode of this size).

Two extrapolations therefore bracket the value at 0.107 A/ μs (Fig. 2):

- **floor:** linear scaling (5.2) from the last published point, with τ held at 55 μs : $Q_{rr} = 2100 \times 0.107/0.7 = 320 \mu\text{C}$;
- **ceiling:** continuation of the straight line of Fig. 4 with slope 0.28: $Q_{rr} = 3000 \times (0.107/3)^{0.28} = 1180 \mu\text{C}$, which corresponds to $\tau \approx 105 \mu\text{s}$.

Both are at $T_{vj} = 150 \text{ }^\circ\text{C}$. The charge scales with the lifetime, and the lifetime grows about linearly with the absolute temperature. So $Q_{rr} \propto T_j^2$. Between 150 and 100 $^\circ\text{C}$ the factor is $(373/423)^2 = 0.78$.

The value chosen for the script, `Qrr_ref = 0.45e-3`, is the geometric mean of the two extremes ($\sqrt{320 \times 1180} = 615 \mu\text{C}$ at 150 $^\circ\text{C}$), moved to 100 $^\circ\text{C}$: $615 \times 0.78 \approx 480 \mu\text{C}$, rounded to 450 μC . The rounding keeps it consistent with set “D” of the charge-controlled model of the earlier notes (0.59 mC at T_{vj} max). It is a defensible middle point, not a measured value. The true value at 0.1 A/ μs can only come from Dynex (§8).

Figure 2 (described). Stored charge of the DS1112SG against the slope of the current, on a log-log plot. The datasheet curve at T_{vj} max is a straight line ($\propto a^{0.28}$) through the rated point (3 A/ μs) and the last published point (0.7 A/ μs). Below 0.7 A/ μs two lines go down to the operating point (0.107 A/ μs , dotted vertical line): the continuation of the straight line (ceiling) and the linear scaling $a\tau^2$ (floor). At the operating point the figure also marks the geometric mean at 150 $^\circ\text{C}$ and the value chosen at 100 $^\circ\text{C}$.

The same plot, rebuilt from the numbers of this note, is [figure 5 of the physics document](#).

5.3 I_{rrm} and t_{rr} from the triangle

Once Q_{rr} and the slope $a = dI_r/dt$ are fixed, the triangle of Fig. 1 ties the other two sizes to the softness $s = t_b/t_a$:

$$I_{rrm} = \sqrt{\frac{2aQ_{rr}}{1+s}}, \quad t_{rr} = \frac{2Q_{rr}}{I_{rrm}}, \quad t_a = \frac{I_{rrm}}{a} < t_{rr} \text{ (PLECS constraint)}. \quad (5.3)$$

At the rated point $s = 1.2$. With a long ramp the tail becomes shorter than the rise. The leftover charge is small, and almost all of it is extracted before the peak. (In the Lauritzen–Ma model the rise lasts $\tau - t_{tail}$ and the tail 2–3 t_{tail} , with $t_{tail} \approx 15\text{--}25\text{ }\mu\text{s}$, that is $s \approx 0.3\text{--}0.5$.) The value $s = 0.4$ is adopted:

$$I_{rrm} = \sqrt{\frac{2 \times 1.07 \times 10^5 \times 0.45 \times 10^{-3}}{1.4}} = 8.3\text{ A}, \quad t_{rr} = \frac{2 \times 0.45 \times 10^{-3}}{8.3} = 109\text{ }\mu\text{s},$$

$$t_a = \frac{I_{rrm}}{dI_r/dt} = \frac{8.3}{1.07 \times 10^5} = 78\text{ }\mu\text{s} < t_{rr} \quad (\text{constraint satisfied}).$$

From here $I_{rrm_ref} = 8.3$ and $T_{rr_ref} = 2 \times Q_{rr_ref} / I_{rrm_ref}$. Table 3 gives the same calculation for the floor and for the ceiling.

Using $s = 1.2$ of the rated point instead of 0.4 lowers I_{rrm} by 20 % for the same charge (5.6 A for the floor) and lengthens t_{rr} . It does not change the energy of the commutation loop in a noticeable way, because that energy depends on I_{rrm}^2 only through the charge.

Table 3: recovery parameter sets at the operating point ($a = 0.107\text{ A}/\mu\text{s}$, $s = 0.4$).

Set	Origin	Q_{rr}	I_{rrm}	t_{rr}	t_a
floor	linear from 0.7 A/ μs , 150 °C	320 μC	7.0 A	92 μs	65 μs
middle (script)	geometric mean, 100 °C	450 μC	8.3 A	109 μs	78 μs
middle at T_{vj} max	geometric mean, 150 °C	615 μC	9.7 A	127 μs	91 μs
ceiling	straight line of Fig. 4, 150 °C	1180 μC	13.4 A	176 μs	125 μs

5.4 Q_{rr} unbalance between the diodes

The diode with the least charge blocks first. It takes the difference ΔQ on its own capacitor. In the model this is represented with one “weak” diode and five nominal ones.

With a long ramp, (5.2) holds and also $I_{rr} \approx a\tau$. So a diode with charge $(1 - u) Q_{rr}$ also recovers with a lower current, $\sqrt{1 - u} I_{rrm}$, and with t_{rr} reduced by the same factor. In the script:

```
Irrm = Irrm_ref*sqrt(1-Max_Qrr_unbalance)
trr   = Trr_ref*sqrt(1-Max_Qrr_unbalance)
Qrr   = Qrr_ref*(1-Max_Qrr_unbalance)      % = 0.5*Irrm*trr (ignored by PLECS)
```

In other words: $Q_{rr} = a\tau^2$ and $I_{rrm} = a\tau$ both come from the same τ . If the charge is 30 % smaller, τ is $\sqrt{0.7} = 0.84$ times smaller, and so are the peak current and the recovery time.

Reducing only t_{rr} while keeping I_{rrm} violates the constraint (5.3): t_a stays at 78 μs while t_{rr} drops to 55 μs for $u = 0.5$.

With five diodes at the top of the band and one at the bottom, u is the full width of the band:

- ≈ 0.2 for a selected set, $\pm 10\text{ }\%$ (banding on Q_S , available from Dynex on request [2]);
- ≈ 0.3 for an unselected lot, $\pm 20\text{ }\%$;
- 0.5–0.6 for mixed lots.

6 Checks on the model

Three quick checks tell whether the parameters were taken in as intended.

1. **Diode waveform at turn-off.** The peak read on the scope must be I_{rrm} , and the integral of the reverse current must be Q_{rr} (8.3 A and 0.45 mC for the middle set). If the values come out very different, the di/dt of the circuit does not match `di_dt_ref` (transformer or current changed) and the model is scaling.
2. **Power in R_{sn} .** With a matched stack, one must find, per resistor,

$$P_{R_{sn}} \approx I_{50}^2 R_{sn} + f \left[\frac{1}{2} \frac{C_{sn}}{n} \frac{V_0^2}{n} + \frac{1}{2} (2L_c) \frac{I_{rrm}^2}{n} \right] = 0.8 + 50 [0.062 + 0.36] \approx 22 \text{ W}, \quad (6.1)$$

where $I_{50} = 58 \text{ mA}$ is the mains-frequency current in C_{sn} . The second term is the charging of the stack to V_0 at every turn-off (it does not depend on R_{sn}). The third term is the energy of the commutation loop at the moment of snap-off, shared between the six resistors. The unbalance adds about $\Delta Q \cdot \hat{v}$ on the first diode, that is 4–8 W for $u = 0.3$. The 26 W read in the simulation are consistent.

3. **Power in R_b .** V_{rms}^2/R_b with $V_{rms} = 0.634 \hat{V}/n = 2063 \text{ V}$ gives 11.2 W at 380 k Ω . The weak diode keeps an offset $\Delta V \approx (n-1)\Delta Q/(nC_{sn})$ of a few hundred volts for the whole blocking period ($R_b C_{sn} = 38 \text{ ms} \gg 13 \text{ ms}$). It adds $2\langle v \rangle \Delta V/R_b$: with $\Delta V = 0.8 \text{ kV}$ about 6.5 W, which explains the 18 W read.

7 Mapping onto the Simscape charge-controlled model (Lauritzen–Ma)

The MATLAB/Simscape twin of the rectifier (model `mv_diode_bridge_design.slx`, folder `matlab_model`) does not use a diode parametrised with Q_{rr} , I_{rrm} and t_{rr} . It uses the component *RR Diode (Lauritzen–Ma)* of the Secom library, file `diode_rr.ssc` of the package `+diode_rr`: a charge-controlled diode [4].

Its parameters are physical: I_s , N , V_t , τ , T_M , R_s , R_r , C_{j0} , V_{jp} , m_g , FC , `dsm`. They must be derived from the same datasheet data of sections 4 and 5. The function `diode_lm_params.m` does this (listing at the end of the section). The mask of the bridge calls it. It receives the structure with the PLECS names (`Vf0`, `rt`, `roff`, `roff_s`, `Cj`, `If0_ref`, `di_dt_ref`, `Qrr_ref`, `Irrm_ref`, plus `Tj` and `N`) and returns the sets D_n for the five nominal diodes and D_u for the unbalanced D11 of each leg. This section derives the formulas that it uses.

7.1 The equations of the block

With v_j the junction voltage and i the diode current:

$$v = v_j + R_s i, \quad q_E = \tau I_s \left(e^{v_j/(NV_t)} - 1 \right),$$

$$\frac{dq_M}{dt} = \frac{q_E - q_M}{T_M} - \frac{q_M}{\tau}, \quad i = \frac{q_E - q_M}{T_M} + \frac{dq_j}{dt} + \frac{v_j}{R_r}. \quad (7.1)$$

Here:

- q_E is the charge that the junction imposes at the edge of the base (exponential law);
- q_M is the charge stored in the base;
- T_M is the transit time with which q_M follows q_E ;
- τ is the carrier lifetime;
- q_j is the SPICE depletion charge (C_{j0} , V_{jp} , m_g , limited by FC);
- R_r is the parallel leakage resistance, that is the `roff` of PLECS.

In static conduction $q_M = \tau i$ and $q_E = (\tau + T_M) i$. With the junction reverse biased, $q_E \rightarrow 0$ and the current is only what the base gives back: $i = -q_M/T_M$.

7.2 From the ramp to the peak and to the tail

With the current forced by the circuit, $i = I_{F0} - a t$, the equation of q_M has the solution

$$q_M(t) = \tau i(t) + a\tau^2 \left(1 - e^{-t/\tau}\right) \approx \tau(i + a\tau) \quad \text{for } t \gg \tau. \quad (7.2)$$

The charge stays behind the current by $a\tau^2$. This is (5.2), used to extrapolate Q_{rr} .

The reverse peak happens when the junction empties, $q_E = q_M + T_M i = 0$, so

$$I_{rrm} = \frac{a\tau^2}{\tau + T_M} = a(\tau - t_{tail}), \quad t_{tail} = \frac{\tau T_M}{\tau + T_M}. \quad (7.3)$$

After the peak, $q_E = 0$, $i = -q_M/T_M$ and $dq_M/dt = -q_M(1/T_M + 1/\tau)$. The current decays exponentially with time constant t_{tail} . The total charge is the triangle of the ramp plus the tail:

$$Q_{rr} = \frac{I_{rrm}^2}{2a} + I_{rrm} t_{tail}. \quad (7.4)$$

Figure 3 compares this waveform with the PLECS triangle for the same Q_{rr} , I_{rrm} and a . The exponential tail, of area $I_{rrm} t_{tail}$, is equivalent to the segment $t_b = 2 t_{tail}$ of the triangle. So the softness implied by the model is $s = 2 t_{tail}/t_a$.

The assumption is the long ramp, $(I_{F0} + I_{rrm})/a \gg \tau$. The ratio is $638/93 \approx 6.9$ for the nominal Dynex set and ≈ 4.5 for the worst case. The script warns if it drops below 3.

7.3 Inversion: from Q_{rr} , I_{rrm} and a to τ and T_M

Equations (7.3) and (7.4) can be inverted in closed form:

$$t_a = \frac{I_{rrm}}{a}, \quad t_{tail} = \frac{Q_{rr}}{I_{rrm}} - \frac{t_a}{2}, \quad \tau = t_a + t_{tail}, \quad T_M = \frac{\tau t_{tail}}{t_a}. \quad (7.5)$$

The fit exists only if $t_{tail} > 0$, that is $Q_{rr}/I_{rrm} > I_{rrm}/(2a)$. With $t_{rr} = 2Q_{rr}/I_{rrm}$ this is exactly $I_{rrm} < t_{rr} a$, the PLECS constraint (5.3). In the script it is an `assert`. So the two libraries ask the same condition of the datasheet data. The choice $s = 0.4$ of §5.3 makes the PLECS triangle and the Lauritzen–Ma tail carry the same charge.

For the nominal Dynex set ($a = 1.07 \times 10^5$ A/s, $Q_{rr} = 450$ pC, $I_{rrm} = 8.3$ A):

$$t_a = \frac{8.3}{1.07 \times 10^5} = 77.6 \text{ } \mu\text{s}, \quad t_{tail} = \frac{450 \times 10^{-6}}{8.3} - \frac{77.6 \times 10^{-6}}{2} = 15.4 \text{ } \mu\text{s},$$

$$\tau = 77.6 + 15.4 = 93.0 \text{ } \mu\text{s}, \quad T_M = \frac{93.0 \times 15.4}{77.6} = 18.5 \text{ } \mu\text{s}.$$

Figure 3 (described). Diode current at turn-off for the nominal Dynex set ($a = 0.107 \text{ A}/\mu\text{s}$, $I_{rrm} = 8.3 \text{ A}$). The PLECS triangle and the response of the charge-controlled model with $\tau = 93 \text{ } \mu\text{s}$, $T_M = 18.5 \text{ } \mu\text{s}$ have the same ramp, the same peak and the same charge. The common ramp is $-at$; the PLECS tail is linear with $t_b = 2t_{tail}$; the Lauritzen–Ma tail is $e^{-t/t_{tail}}$. Both carry $Q_{rr} = 450 \text{ } \mu\text{C}$.

The same figure, computed from these parameters, is [figure 2 of the physics document](#).

The unbalance $Q_{rr}(1-u)$, $I_{rrm}\sqrt{1-u}$ of §5.4 scales both Q_{rr}/I_{rrm} and I_{rrm}/a by the same factor $\sqrt{1-u}$. So t_a , t_{tail} , τ and T_M of the weak diode are all $\sqrt{1-u}$ times the nominal ones ($\times 0.837$ for $u = 0.3$).

7.4 Conduction: I_s , N and R_s

In static conditions (7.1) gives

$$i = \frac{\tau}{\tau + T_M} I_s \left(e^{v_j/(NV_t)} - 1 \right), \quad v = v_j + R_s i.$$

The PLECS straight line, $V_{f0} + r_t i$, is made to coincide with the model both in value and in slope at $i = I_{F0}$, with $V_t = kT_j/q = 32.16 \text{ mV}$ at $T_j = 100 \text{ }^\circ\text{C}$ and $N = 1$:

$$R_s = r_t - \frac{NV_t}{I_{F0}}, \quad v_j(I_{F0}) = V_{f0} + NV_t, \quad I_s = \frac{I_{F0}}{e^{v_j/(NV_t)} - 1} \cdot \frac{\tau + T_M}{\tau}. \quad (7.6)$$

The slope of the exponential at I_{F0} , NV_t/I_{F0} ($0.54 \text{ m}\Omega$), takes up more than half of r_t . So R_s is small ($0.394 \text{ m}\Omega$ for Dynex, $0.334 \text{ m}\Omega$ for Infineon), and N cannot exceed $r_t I_{F0}/V_t \approx 1.7$ without making R_s negative.

I_s depends on the recovery set only through $(\tau + T_M)/\tau$, which is the same for D_n and D_u : $1.85 \times 10^{-11} \text{ A}$ for Dynex ($1.88 \times 10^{-11} \text{ A}$ in the worst case), 1.15×10^{-10} to $1.38 \times 10^{-10} \text{ A}$ for Infineon.

Far from I_{F0} the exponential curve moves away from the straight line. But the losses in the diodes do not enter the sizing of the resistors.

7.5 The other parameters

- $R_r = \text{roff}$ for the unbalanced diode (D11, like Drr1 in PLECS) and $R_r = \text{roff_s}$ for the other five: 1.2 and $2.1 \text{ M}\Omega$ for Dynex (§4.3). For Infineon, with no spread data, $\text{roff_s} = \text{roff} = 2.9 \text{ M}\Omega$.
- $C_{j0} = \text{Cj} = 0.5 \text{ nF}$ with $m_g = 0$, that is a constant capacitance as in PLECS.
- $V_{jp} = 0.7 \text{ V}$, $FC = 0.5$ and $\text{dsm} = 10 \text{ mV}$ are default values that do not affect the results.
- V_{RRM} does not enter the block. The voltage margin must be checked on the waveforms, as in PLECS.

7.6 The twelve sets

Table 4 collects the recovery parameters for the two candidate diodes and the three cases, with $u = 0.3$. In `init_model.m` they are chosen with `diode_type`, `diode_case` and `Max_Qrr_unbalance`.

The mask of the bridge calls `diode_lm_params`, and the 36 diodes read `Dn.*` or `Du.*`. The Infineon D711N65T sets ($V_{f0} = 0.84$ V, $r_t = 0.87$ m Ω) come from the note `resistor_thermal_note.md`, §7.

Table 4: Lauritzen–Ma parameters of the twelve sets ($a = 1.07 \times 10^5$ A/s, $I_{F0} = 60$ A, $u = 0.3$, $T_j = 100$ °C, $N = 1$). D_n : nominal diodes; D_u : unbalanced D11 ($Q_{rr} \times 0.7$, $I_{rrm} \times 0.837$).

Diode	Case		Q_{rr} (μ C)	I_{rrm} (A)	t_{tail} (μ s)	τ (μ s)	T_M (μ s)	I_s (A)
Dynex	nominal	D_n	450	8.30	15.43	93.00	18.50	1.85×10^{-11}
		D_u	315	6.94	12.91	77.81	15.48	1.85×10^{-11}
	worst	D_n	1200	13.40	26.94	152.17	32.73	1.88×10^{-11}
		D_u	840	11.21	22.54	127.31	27.38	1.88×10^{-11}
	floor	D_n	320	7.00	13.00	78.42	15.59	1.85×10^{-11}
		D_u	224	5.86	10.88	65.61	13.04	1.85×10^{-11}
Infineon	nominal	D_n	300	6.60	14.61	76.30	18.08	1.23×10^{-10}
		D_u	210	5.52	12.23	63.83	15.12	1.23×10^{-10}
	worst	D_n	590	9.80	14.41	106.00	16.68	1.15×10^{-10}
		D_u	413	8.20	12.06	88.68	13.95	1.15×10^{-10}
	floor	D_n	190	4.80	17.15	62.01	23.71	1.38×10^{-10}
		D_u	133	4.02	14.35	51.88	19.84	1.38×10^{-10}

7.7 Check

Integrating (7.1) numerically with the current forced (nominal Dynex set), q_E reaches zero at 638.3 μ s, with $I_{rrm} = 8.300$ A and $Q_{rr} = 450.0$ μ C. So (7.5) is exact within the long-ramp assumption. (The script `lm_params_check.py` in the folder repeats the calculation and the integration for all the sets.)

In the Simscape test circuit (driven source with $L = 62$ mH, rise to 60 A in 1 ms and then $di/dt = -1.07 \times 10^5$ A/s, diode with $R_{sn} + C_{sn}$ in parallel, solver `daessc`, maximum step 0.1 μ s), the first negative lobe of the diode current gives the values of Table 5. I_{rrm} is within 2 % (the -1.7 % of the Dynex worst case is the ramp being shorter with respect to τ). Q_{rr} is within 5 % and always in excess, because the lobe also contains the current of C_j and R_r .

In the full bridge (Dynex, $u = 0.3$, grid +10 %, steady state 80–100 ms), the commutation di/dt is 1.14×10^5 A/s. The diodes recover with $I_{rrm} = 8.15$ A, $Q_{rr} = 458$ μ C (D12–D16) and 7.48 A, 326 μ C (D11). So the losses in the resistors can be compared directly with those of PLECS (§6).

Table 5: Simscape test circuit at $di/dt = 1.07 \times 10^5$ A/s: measured value / target value of the set.

Diode	Case	D_n : I_{rrm} (A)	D_n : Q_{rr} (μ C)	D_u ($u = 0.3$): I_{rrm} (A)	D_u : Q_{rr} (μ C)
Dynex	nominal	8.29 / 8.30	456 / 450	6.94 / 6.94	323 / 315
	worst	13.17 / 13.40	1174 / 1200	11.12 / 11.21	837 / 840
	floor	7.00 / 7.00	327 / 320	5.86 / 5.86	232 / 224
Infineon	nominal	6.60 / 6.60	307 / 300	5.52 / 5.52	217 / 210
	worst	9.77 / 9.80	594 / 590	8.19 / 8.20	420 / 413
	floor	4.80 / 4.80	197 / 190	4.02 / 4.02	140 / 133

7.8 Listing of diode_lm_params.m

```
function [Dn, Du, info] = diode_lm_params(d, u)
% d: struct from init_model.m (PLECS names): Vrrm, roff, roff_s, Cj, Vf0, rt,
%   If0_ref, di_dt_ref, Qrr_ref, Irrm_ref, Tj, N
% u: Max_Qrr_unbalance. Dn, Du: RR Diode (Lauritzen-Ma) parameters
%   for the nominal diodes and for the unbalanced one.
if nargin < 2, u = 0; end
assert(u >= 0 && u < 1, 'Max_Qrr_unbalance must be in [0, 1).');
k_q = 1.380649e-23/1.602176634e-19;
Vt = k_q*(d.Tj + 273.15);
N = d.N;
a = d.di_dt_ref;
Dn = one_set(d, d.Qrr_ref, d.Irrm_ref, d.roff_s);
Du = one_set(d, d.Qrr_ref*(1 - u), d.Irrm_ref*sqrt(1-u), d.roff);
info.Vt = Vt; info.Trr_n = 2*Dn.Qrr/Dn.Irrm; info.Trr_u = 2*Du.Qrr/Du.Irrm;
function P = one_set(d, Qrr, Irrm, Rr)
    trr = 2*Qrr/Irrm;
    assert(Irrm < trr*a, ['Irrm = %.3g A >= trr*di/dt = %.3g A: ' ...
        'no Lauritzen-Ma fit (t_tail <= 0).'], Irrm, trr*a);
    t_tail = Qrr/Irrm - Irrm/(2*a);
    tau = t_tail + Irrm/a;
    TM = tau*t_tail*a/Irrm;
    if (d.If0_ref + Irrm)/a < 3*tau
        warning('diode_lm_params:ramp', ['Short ramp: (If0+Irrm)/a = ' ...
            '%.3g s vs tau = %.3g s; Irrm/Qrr will deviate.'], ...
            (d.If0_ref + Irrm)/a, tau);
    end
    Rs = d.rt - N*Vt/d.If0_ref;
    assert(Rs > 0, 'rt too small for N = %g: lower N.', N);
    vj = d.Vf0 + N*Vt;
    Is = d.If0_ref/(exp(vj/(N*Vt)) - 1)*(tau + TM)/tau;
    P = struct('Is', Is, 'N', N, 'Vt', Vt, 'tau', tau, 'TM', TM, ...
        'Rs', Rs, 'Rr', Rr, 'Cj0', d.Cj, 'Vjp', 0.7, 'mg', 0, ...
        'FC', 0.5, 'dsm', 10e-3, 'Qrr', Qrr, 'Irrm', Irrm, ...
        'trr', trr, 't_tail', t_tail);
end
end
```

The comments inside the listing were translated into English; the code is unchanged.

8 Data to ask Dynex for

The same three pieces of data close both the PLECS set and, through (7.5) and (7.6), the Simscape set.

- Q_S and I_{rr} measured at 0.1 A/ μ s, 60 A, $T_{vj} = 100$ and 125 °C (or the ambipolar lifetime of the wafer). They fix Q_{rr_ref} and I_{rrm_ref} between the floor and the ceiling of Table 3, that is the power of R_{sn} between 12 and 50 W.
- Reverse current at 3.3 kV for T_j between 50 and 125 °C, or the banding on leakage. It fixes $roff$ and the maximum T_j allowed by R_b .
- Availability and width of the banding on Q_S . It fixes $Max_Q_{rr_unbalance}$.

References

1. Dynex Semiconductor, “DS1112SG rectifier diode,” data sheet DS4181-5.0, August 2001.
2. Dynex Semiconductor, “Series and parallel connection of thyristors and diodes,” Application Note AN6531.
3. Plexim GmbH, “Diode with Reverse Recovery,” PLECS User Manual 5.0, component reference.
4. P. O. Lauritzen and C. L. Ma, “A simple diode model with reverse recovery,” IEEE Trans. Power Electron., vol. 6, no. 2, pp. 188–191, 1991.